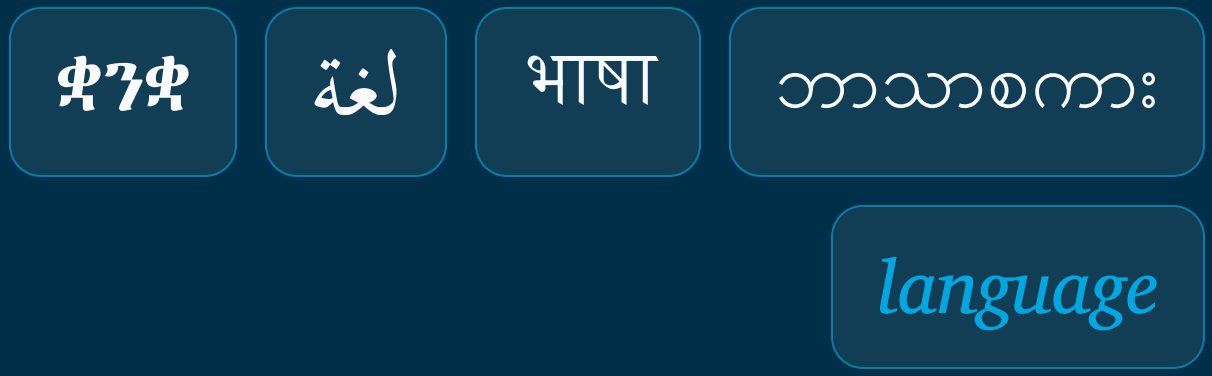
# Motif in Brief

A working demonstration of tools for FieldWorks grammars, built on forty years of SIL parsing work.

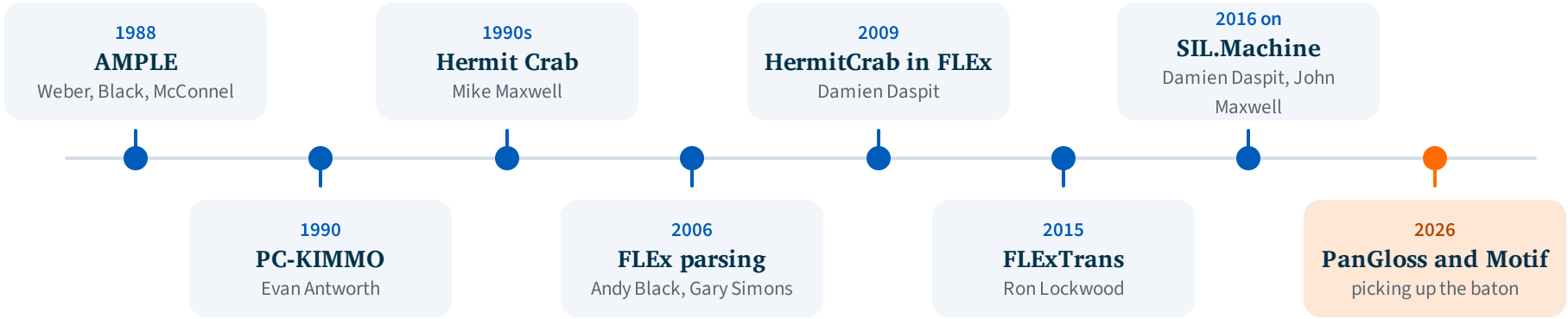



SIL Global · HermitCrab, PanGloss and Motif · October 2026

Where this comes from

# Forty years toward this

A **grammar** describes how the words of a language are built. A **parser** runs it backwards: give it a word, and it tells you what the word is made of. SIL has been building both for decades.




HermitCrab's maintainers continue its development today, with 5–10× speedups from algorithm work in SIL.Machine.

## Grammars being built now

FLExTrans helps draft Scripture in the Philippines and Peru. Parser workshops in Manila (2025) and Nairobi (2026) train linguists to build grammars for their own languages. Many more grammars sit in FieldWorks projects, some mature and many "getting there", built patiently over years.

PICKING UP THE BATON Motif does not start over. It keeps every grammar already built, and tries out tools that help the people building them go further.

The need

# What builders meet as a grammar grows

The linguistics is sound. What gets harder as a grammar grows is everything around it.

### Waiting

Some combinations of rules become expensive to search. On one Amharic grammar, 7,000 words take about half an hour, and one word in six hits a five-second limit. When re-parsing is slow, it happens less, and the grammar improves more slowly.

### Seeing

Try a Word shows how one word was parsed. Nothing shows, across a whole text, where the time goes, which rules never fire, or which let in forms the language does not have.

### Reaching

A finished grammar already does real work in FieldWorks and FLExTrans, and people publish with it. Paratext, word processors and keyboards cannot yet use it for back-translations or spelling.

**As a grammar grows today**

change | wait: minutes to hours | check

**The goal** — A target, not a measurement

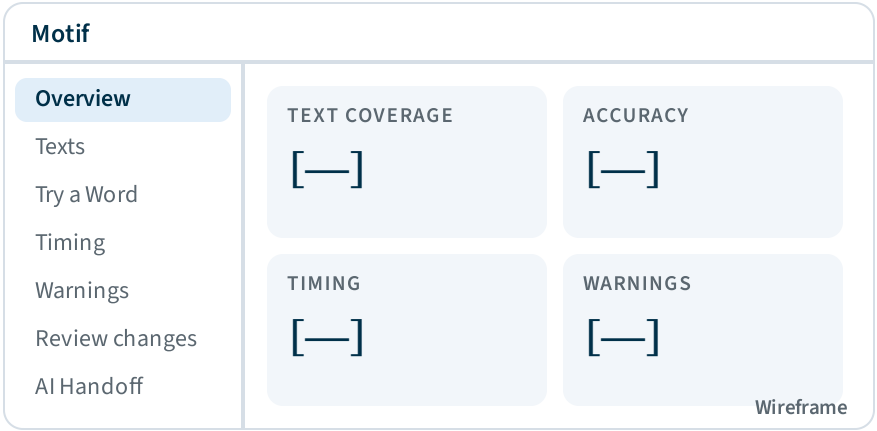
change | re-parse: seconds | check | change | re-parse: seconds | check | change | re-parse: seconds | check

WHAT WOULD HELP Faster re-parsing, a clear view of what the grammar is doing, and a safe way to try a change and see whether it helped.

# Motif today: a working demonstration

**Motif** is a desktop and command-line demonstration for the FieldWorks project a team already uses. Invited testers have a Windows installer, with macOS and Linux previews. It opens projects and backups; large Texts and word lists load as you scroll.

## The main pages in the window




- **Overview**: how much of the chosen texts the grammar parses, how accurately, and how fast
- **Texts**: every word in context, parsed or not
- **Try a Word**: the parser's reasoning for any word, step by step, including where it gave up
- **Timing**: the slowest words and the rules that cost the most
- **Warnings**: grammar-health problems to fix in FieldWorks
- **Review changes**: check marked changes, then apply them to the FieldWorks project
- **AI Handoff**: the whole picture, packaged for an AI assistant

## Changes checked before they land

An author marks analysis changes in Texts. Motif tries them on a copy of the project (a **Dry Run**), re-measures the grammar, and applies them to the FieldWorks project in one step, with a receipt. It will not apply while FieldWorks has the project open.

## The AI Handoff

One click writes the grammar, selected texts, available results and diagnostics, a helper script and a short guide. The author chooses what to share with an AI chat, which can explain why a word failed. Motif does not send the files.

STILL A DEMONSTRATION The beta measures and stages analysis changes. Advanced AI work is taking shape in current source; daily use and FieldWorks integration need pilot teams.

How it works

# The grammar-improvement loop

PanGloss measures, Motif shows, an AI assistant drafts, and a person decides. Analysis changes already follow this loop. In current development, connected assistants can also draft, try and revise supported grammar changes before handing them over.




### What the author needs

A good understanding of their language, and the judgement to say whether a proposed change is right.

### What makes it fast

Parallel parsing; diagnostics that point at the problem; an assistant that reads traces so the author does not have to.

### What keeps it honest

The grammar is tuned, never the parser. Changes are tried on a copy; the assistant cannot Apply. A person decides what lands.

Coming next

# Coming next: from advice to checked changes

Work taking shape beyond the released beta, with the same rule throughout: the linguist decides.

### Connect an assistant

Advanced AI mode lets Claude or Codex read Known projects and draft supported grammar changes as Proposals. The assistant can Dry Run, Trial, read the differences, revise and Finalize. It cannot Apply.

### Find grammar worth tidying

Parsimony points to duplication, broad conditions and listed alternates that a sound rule might explain. Each finding names its evidence and limits. A linguist can keep it, ask a question or defer it. Recipes guide supported fixes through checked Proposals; the page never repairs the grammar itself.

### Review what would change

A Proposal's Trial measures a copy of the project. The evidence and word-card work is in place; a complete window showing today beside the Proposal is still being finished.

NEXT PROOF Try the whole workflow with pilot teams: do the suggestions preserve valid readings, avoid forbidden ones, and help people improve their grammars?

Underneath

# PanGloss underneath

**PanGloss** is a port of HermitCrab to Rust. It follows the same algorithms and shares their improvements, and it is what Motif runs.

## Faithful to HermitCrab

Grammar fixtures that trace the contours of the parser hold PanGloss to HermitCrab's answers. Where the two differ, the difference is written up and tracked. It reads FieldWorks projects directly. Documented differences and unsupported patterns remain visible; compatibility is checked, not assumed.

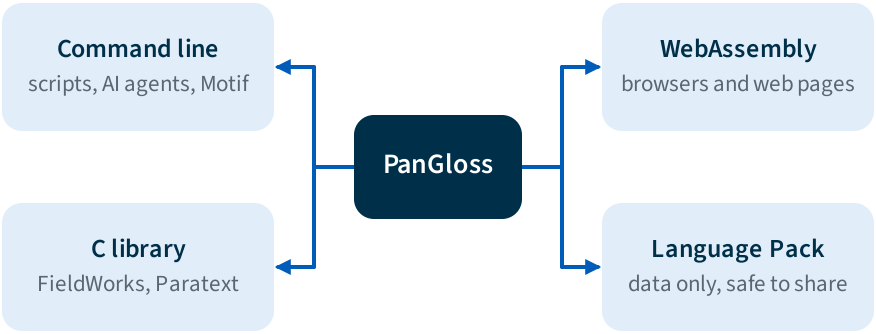
## Faster, never by guessing

Better algorithms are shared with HermitCrab and SIL.Machine. Parallel batches now spread words across several threads; Motif uses at most half the machine. Difficult single words can still dominate the wait.

## Measures across a whole word list

- **Timing** for every word, with runaway words stopped and reported
- **Statistics**: where the time goes, and which rules never fire
- **Grammar health**: structural problems and stored analyses lost to authored rules; unfinished checks are reported

## Native releases; more hosts ahead




WHY A PORT Every hour already spent on a FLEx grammar is kept, and every answer can be checked against the original.

Where it goes

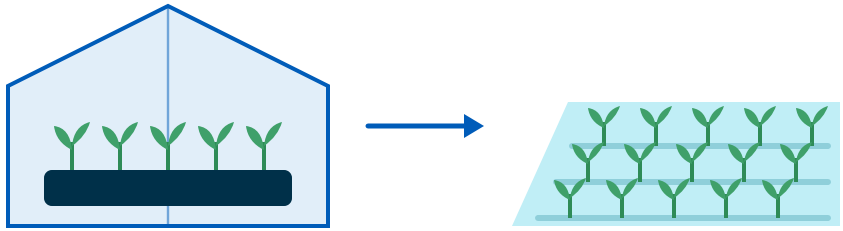
# From demonstration to FieldWorks

Motif is not meant to be one more program to install. It is a demonstration where new ideas meet real grammars quickly, without putting a stable, widely used FieldWorks at risk. What works goes home to FieldWorks.

## Phase 1: into FieldWorks

- PanGloss as a parser FieldWorks can call
- Assessment, the Overview, Try a Word, Timing and Warnings
- Checking and applying changes, with a Dry Run first
- The AI Handoff

Motif is built on .NET 10 and Avalonia, the same move FieldWorks' own interface work is making, so a later merge is a move, not a rewrite.



**Motif** · the nursery

**FieldWorks** · the field

## Phase 2: perhaps its own app

Learning from parallel texts and harvesting words statistically needs large bodies of outside text that do not belong in a FieldWorks project. Whether that lives in FieldWorks or in a companion app is still open.

## Beyond FieldWorks

Once grammars are fast and easy to build, they can work in **Paratext**, for back-translations and spelling checks, and later in keyboards, browsers and word processors. The longer paper, *A Working Grammar for Every Language*, sets out that path.

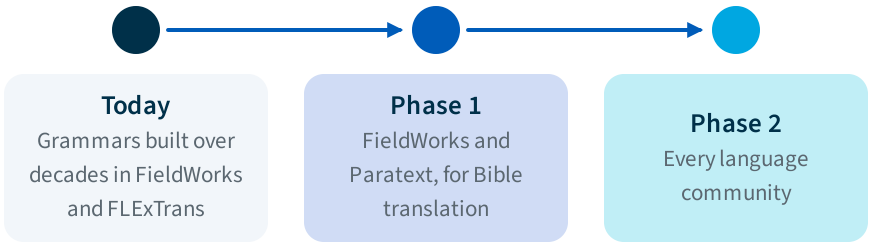
Next

# What we would like to do next

1. **Plan together** with the FieldWorks and HermitCrab maintainers, so speedups and tools flow both ways.
2. **Choose pilot teams**: five to ten translation projects with a FieldWorks lexicon, interlinear texts and a grammar under way.
3. **Publish a baseline** of today's parse speed and Text Coverage on their data.
4. **Prove the loop with pilot teams**, from the assistant's advice to a checked Proposal and a person's Apply.
5. **Bring it home**: move what Motif has demonstrated into FieldWorks.


### With thanks

This work stands on that of David Weber, Andy Black, Stephen McConnel, Evan Antworth, Mike Maxwell, Gary Simons, Damien Daspit, John Maxwell and Ron Lockwood, and on every linguist who has built a grammar in FieldWorks.


Every Phase 1 grammar is a Phase 2 grammar waiting to be published.

THE INVITATION Help carry forty years of SIL's parsing work the rest of the way: into FieldWorks first, then to every translation team, and one day to every language community.